

# Jia-Yu Chen

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## EDUCATION

### University of Michigan GPA: 3.6/4.0

Ann Arbor, MI

*Bachelor of Science in Engineering, Electrical Engineering*

*Aug. 2023 – Dec. 2026*

- Relevant Coursework: VLSI Design, Embedded Control Systems, Analog & Digital IC Design, Logic Design, Signals & Systems, Electromagnetics, Semiconductor Devices, Data Structures & Algorithms

## EXPERIENCE

### Electrical Team Lead — NASA SunRISE Mission

Jan. 2025 – Present

- Led a 10-person electrical subteam within a 30-member HelioPhysics Ground Lab; coordinating interferometry, GPS timing and positioning, and calibration efforts with weekly execution meetings and Slack-based task tracking
- Supported SunRISE space-weather objectives by advancing a Michigan-based Long-Wavelength Array (LWA) ground station collecting solar radio data across 5–85 MHz for burst detection and downstream ML classification
- Configured field deployment at Peach Mountain Observatory, captured raw .T00 datasets, and defined post-processing steps to extract inter-antenna timing and geometry differences
- Produced installation documentation for 26 partner high schools for antenna kits assembly

### BMS & Dashboard Designer — Solar Car Team – Microsystems

Sep. 2023 – Apr. 2025

- Designed and validated a compact power-input protection circuit to prevent PCB failures from reverse polarity wiring and overcurrent/overvoltage events on 12 V systems
- Selected and sized components (series Schottky diode, TVS diode, and resettable fuse) to balance protection and efficiency; justified parts via forward-drop, standoff voltage, and clamping behavior
- Built schematics/PCB in Altium Designer; fabricated and hand-assembled prototypes; bench-tested reverse supply and surge/current injection for fault tolerance
- Measured system impact, demonstrating <1 W added dissipation per board with acceptable thermal rise; drove adoption as a standard feature in future PCBs and reduced BMS sensing board footprint by 25%.

## PROJECTS

### Layout Designer — EECS 427 RISC-V Processor Design

Jan. 2026 – Apr. 2026

- Designed a two-stage pipelined baseline RISC-V processor including register file, ALU, barrel/log shifter, program counter, and control unit, delivering schematic + layout in Cadence using the TSMC standard-cell library.
- Implemented a high-speed Kogge–Stone adder-based ALU; used floorplanning, stick diagrams, and HSPICE speedpath analysis to optimize transistor sizing for PPA(power/performance/area) with complicated wiring
- Drove synthesis + place-and-route (APR) for PC/controller in Cadence Innovus, editing .tcl scripts, achieving 90% core density, planning I/O pin placement, and closing timing with zero setup/hold violations post-route
- Built a 16×16 multiplier using a systolic array + Wallace-tree compression, leveraging a hierarchical APR flow with black-boxed modules, and verified physical correctness with DRC/LVS while minimizing layout area.

### Student Electrical Engineer — Sartorius Hardware Incucyte Project

Jan. 2026 – Present

- Contributed to an incubator subsystem integration for a live-cell imaging system (37°C ± 0.5°C and 95% RH).
- Defined controls implementation approach (sensor acquisition over I2C, control on Arduino/Raspberry Pi, PWM to MOSFET actuation) to support future PID tuning and integrated testing
- Designed experiments using an 8-point temperature/humidity sensor grid at the cell-tray plane to assess uniformity and transient response across configurations

### EECS 461: Embedded Cruise Control System Design

Sep. 2025 – Dec. 2025

- Developed a PI closed-loop cruise controller in MATLAB/Simulink; tuned gains to meet speed-tracking and stability requirements from driver inputs (switches and potentiometers)
- Implemented CAN communication with interrupt-driven message handling, byte unpacking, and signal demultiplexing; validated end-to-end control behavior through bench testing

## TECHNICAL SKILLS

**Hardware:** PCB Design (Altium), Analog/Digital Circuits, Power Electronics (TVS, Schottky, Fusing), BMS, RF & Antenna Systems, Signal Integrity, Oscilloscope, SMD Soldering

**Embedded Systems:** C/C++, Microcontrollers (Arduino, Raspberry Pi), I2C, PWM, CAN Bus, Interrupts

**Programming:** Python, MATLAB, C++, Java, SQL